

1 Грамматика

```
letter = "a" | "b" | ... | "z" | "A" | "B" | ... | "Z" ;  
digit = "0" | "1" | ... | "9" ;  
newline = "\n" ;  
identifier = letter | letter , id_tail ;  
id_tail = symbol | symbol , id_tail ;  
symbol = letter | digit | "_" ;  
program = statement_list | ;  
statement_list = statement | statement , statement_list ;  
statement = let_binding | term_stmt ;  
term_stmt = term , newline ;  
let_binding = "let" , identifier , "=" , term , newline ;  
term = lambda_expr | application ;  
application = operand | operand , app_tail ;  
app_tail = operand | lambda_expr | operand , app_tail ;  
lambda_expr = "\" , id_list , "." , term ;  
id_list = identifier | identifier , id_list ;  
operand = identifier | "(" , term , ")" ;
```

2 Пример1

```
\x y.x \n
```

1. program -> statement_list
2. statement_list -> statement
3. statement -> term_stmt
4. term_stmt -> term , newline
5. term -> lambda_expr
6. lambda_expr -> "\" , id_list , "." , term
7. id_list -> identifier(x) , id_list
8. id_list -> identifier(y)
9. term -> application
10. application -> operand
11. operand -> identifier(x)
12. newline -> "\n"

\x y.x \n

3 Пример2

let I = \x.x \n
I z \n

1. program -> statement_list
2. statement_list -> statement , statement_list
3. statement -> let_binding
4. let_binding -> "let" , identifier(I) , "=" , term , newline
5. term -> lambda_expr
6. lambda_expr -> "\" , id_list , "." , term
7. id_list -> identifier(x)
8. term -> application
9. application -> operand
10. operand -> identifier(x)
11. newline -> "\n"
12. statement_list -> statement
13. statement -> term_stmt
14. term_stmt -> term , newline
15. term -> application
16. application -> operand , app_tail
17. operand -> identifier(I)
18. app_tail -> operand
19. operand -> identifier(z)
20. newline -> "\n"

```
let I = \x.x \n I z \n
```